Comparative Study of Turbulent Pipe-Flow Models

Evaluation of radial eddy-viscosity closures using velocity profiles and friction factors

1. Aim

To compare different radial eddy-viscosity models for fully developed turbulent pipe flow. The comparison is based on mean velocity profiles, profile fullness, Darcy friction factor and the corresponding implications for wall shear and pressure drop.

2. Computational Model and Parameters

The flow is treated as steady, fully developed and axisymmetric. A radial finite-volume momentum solver is used, with the pressure gradient adjusted to maintain a unit bulk velocity. The models considered are laminar flow, mixing length, Van-Driest-damped mixing length, equilibrium k-epsilon and an equilibrium k-omega/SST surrogate.

Parameter	Value
Coordinate system	Axisymmetric cylindrical
Radial grid points, nx	160
Axial grid points, ny	61
Reynolds-number range	10^4 to 10^5
Number of Reynolds-number cases, N_Re	3
Maximum iterations	1500
Convergence tolerance	1 × 10^(-7)
Bulk velocity, U_b	1 (prescribed)

3. Governing Equation and Boundary Conditions

For fully developed axisymmetric flow, the axial momentum balance is governed by the pressure gradient and the radial gradient of total shear stress. The effective viscosity contains the molecular kinematic viscosity and the modelled turbulent eddy viscosity. The closures differ in the expression used for the turbulent viscosity, nu_t.

Axis symmetry is imposed at the pipe centreline and the no-slip condition is applied at the wall. The pressure gradient is adjusted until the prescribed unit bulk velocity is obtained.

4. Numerical Procedure and Turbulence Closures

The radial momentum equation is discretized using a finite-volume approach. The laminar case provides the parabolic Poiseuille reference profile. Turbulent cases use eddy-viscosity closures: the undamped mixing-length model, Van-Driest damping near the wall, an equilibrium k-epsilon model and a k-omega/SST surrogate.

5. Quantities Used for Comparison

The main comparison measures are centreline velocity, the bulk-to-centreline velocity ratio as an indicator of profile fullness, Darcy friction factor, and the associated wall shear and pressure drop. Friction factors are compared with the Blasius and smooth-pipe Haaland reference correlations.

6. Numerical Results

The source report provides tabulated results at Reynolds numbers 10^4 and 10^5. Although three Reynolds-number cases are specified in the setup, the intermediate case near 3.16 × 10^4 is not included in the supplied result table or figures.

Quantity	Re = 10^4	Re = 10^5
Blasius friction factor	0.0316	0.0178
Haaland friction factor	0.0309	0.0178
Laminar U_cl (U_b/U_cl)	2.00 (0.50)	2.00 (0.50)
Mixing length U_cl (fullness)	1.25 (0.80)	1.14 (0.88)
Van Driest U_cl (fullness)	1.20 (0.83)	1.14 (0.88)
k-epsilon U_cl (fullness)	1.165 (0.86)	1.115 (0.90)
k-omega/SST U_cl (fullness)	1.20 (0.83)	1.08 (0.93)
Laminar Darcy factor	0.0064 (0.20 × Blasius)	0.00064 (0.04 × Blasius)
Mixing-length Darcy factor	0.083 (2.6 × Blasius)	0.021 (1.20 × Blasius)
Van-Driest Darcy factor	0.038 (1.19 × Blasius)	0.019 (1.09 × Blasius)
k-epsilon Darcy factor	0.026 (0.81 × Blasius)	0.0165 (0.93 × Blasius)
k-omega/SST Darcy factor	0.014 (0.45 × Blasius)	0.0062 (0.35 × Blasius)

7. Velocity-Profile Fullness

All turbulent closures produce a fuller velocity profile than the parabolic laminar profile. At Re = 10^4, the bulk-to-centreline ratios are approximately 0.80 for mixing length, 0.83 for Van Driest, 0.86 for k-epsilon and 0.83 for the k-omega/SST surrogate. By this measure, k-epsilon gives the fullest profile at this Reynolds number. At Re = 10^5, the corresponding ratios are 0.88, 0.88, 0.90 and 0.93, making the k-omega/SST surrogate the fullest by this measure.

The source notes that the k-omega/SST profile has a particularly flat core and a sharp near-wall drop. The profiles become fuller as Reynolds number increases. The mixing-length and Van-Driest profiles are nearly identical at Re = 10^5, since wall damping affects only a thin near-wall region at this higher Reynolds number.

8. Friction-Factor Comparison

The Blasius reference values are 0.0316 and 0.0178 at Re = 10^4 and 10^5, while the Haaland values are 0.0309 and 0.0178. Among the turbulent closures, Van Driest is closest to these references: about 19% above Blasius at Re = 10^4 and 9% above at Re = 10^5. The k-epsilon values are slightly below the references. The undamped mixing-length model over-predicts friction, particularly at Re = 10^4, whereas the k-omega/SST surrogate substantially under-predicts it.

At fixed bulk velocity, wall shear and pressure drop are proportional to the friction factor. They therefore follow the same relative ordering and discrepancies as the reported Darcy friction factors.

9. Consistency and Limitations

The source report points out that profile fullness and friction factor do not agree consistently for every closure. For example, the mixing-length friction factor at Re = 10^4 would imply a different centreline velocity from the value read from its computed profile. Similarly, the k-omega/SST surrogate gives the fullest profile at Re = 10^5 but the lowest friction factor. This suggests that the wall-shear calculation is sensitive to near-wall treatment, including the first-cell velocity gradient or a wall-function approach.

The k-epsilon and k-omega profiles end at non-zero velocity at the wall coordinate r/R = 1, suggesting that their first sampled point may lie outside the viscous sublayer. Their friction factors should therefore be interpreted cautiously. The friction-factor plots show one Reynolds number each; the Blasius and Haaland references appear in the legends but are

not drawn as curves. The intermediate Reynolds-number case, wall shear, pressure drop and maximum turbulent-viscosity ratio are not separately reported in the supplied figures.

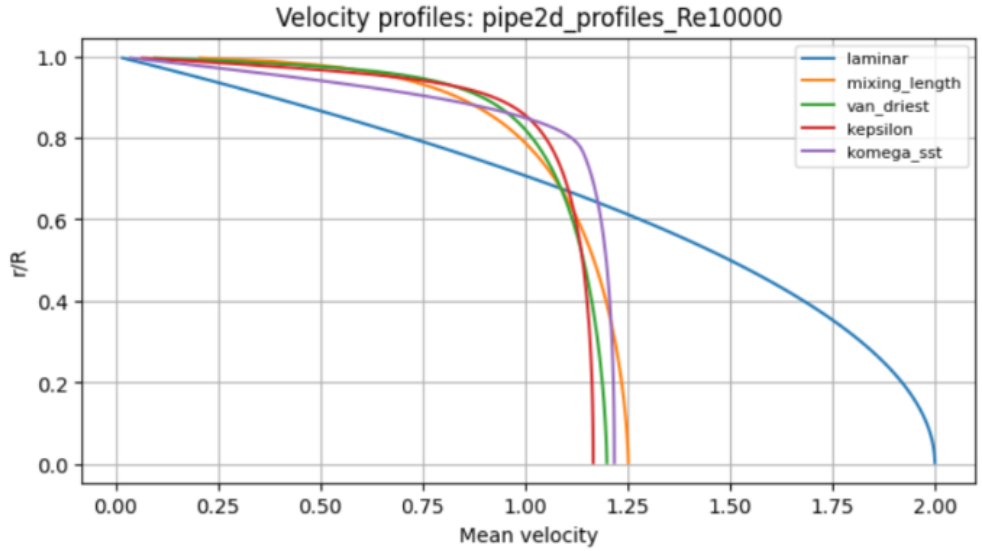
10. Conclusion

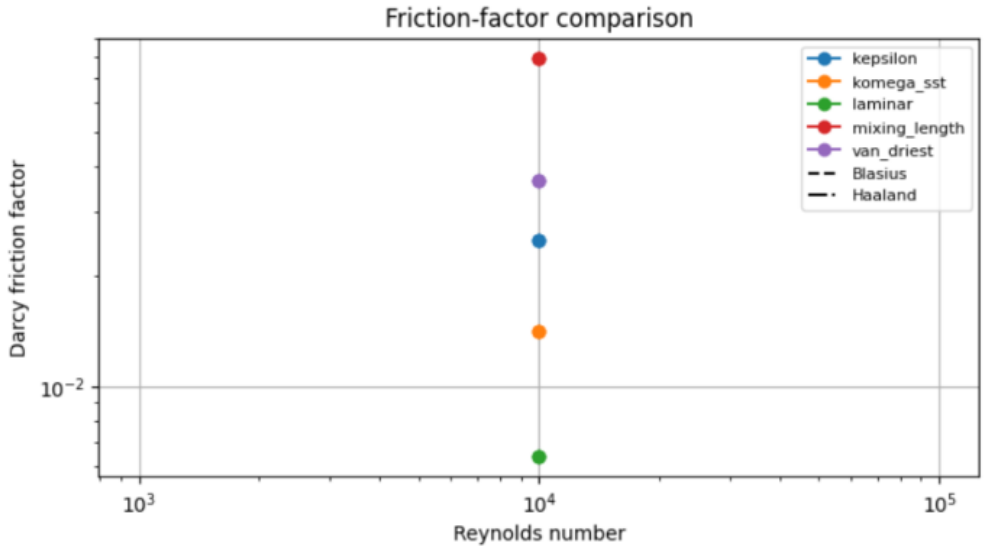
The comparison shows that all turbulent closures produce fuller-than-laminar velocity profiles, and that the profiles become fuller as Reynolds number increases. The Van-Driest-damped mixing-length model gives friction factors closest to the Blasius and Haaland references among the tested closures. The k-epsilon model is somewhat below the correlations, the undamped mixing-length model over-predicts friction, and the k-omega/SST surrogate under-predicts it substantially.

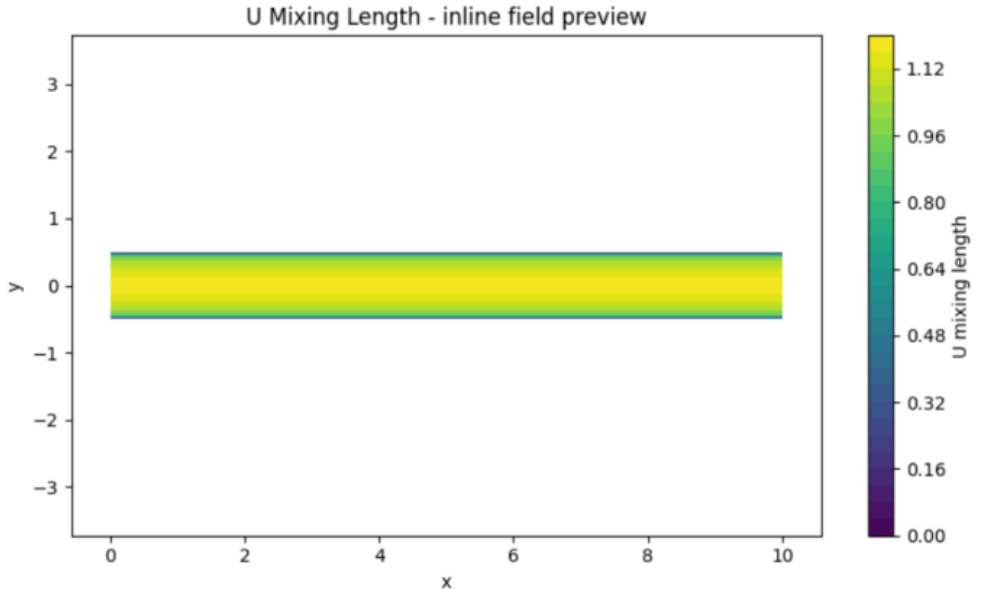
A fuller velocity profile does not necessarily indicate a more accurate friction prediction. The disagreement between profile shape and wall-friction results highlights the importance of near-wall resolution and wall treatment. A more complete assessment would include the intermediate Reynolds-number case, explicit reference curves, first-cell y-plus information, wall-shear and pressure-drop values, and the maximum turbulent-viscosity ratio.

11. Original Simulation Figures

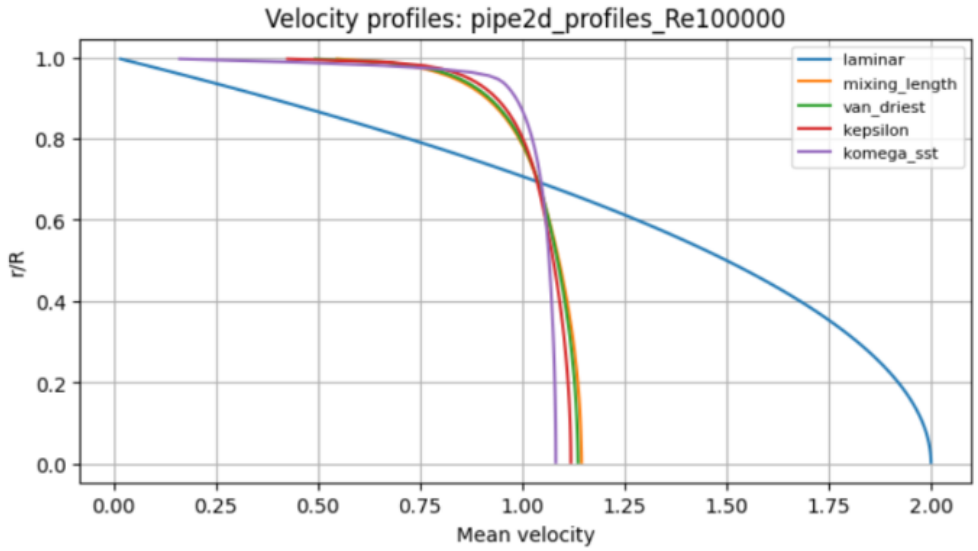
The original velocity profiles, friction-factor comparisons and mixing-length field visualizations are reproduced in the following pages.

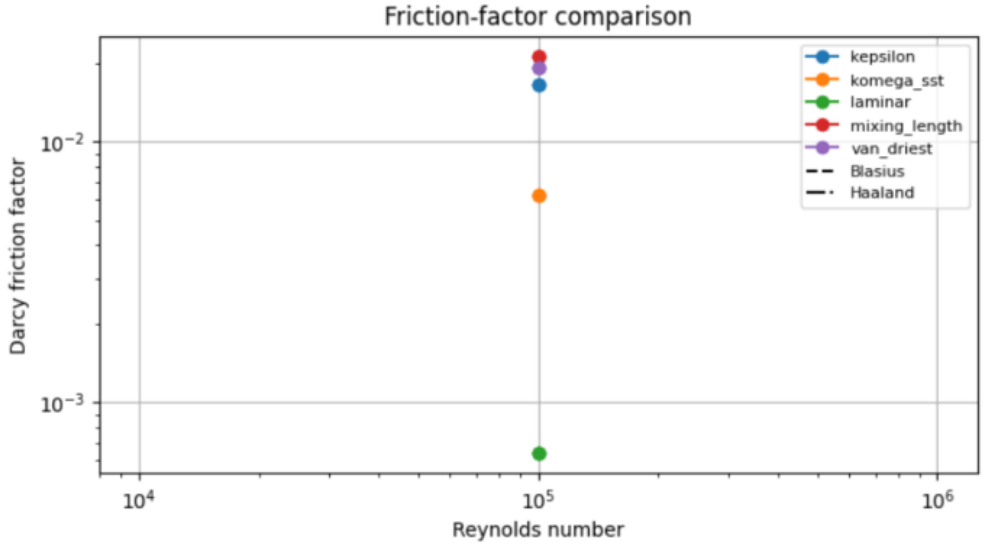

Velocity profiles: pipe2d_profiles_Re10000
laminar
mixing_length
van_driest
kepsilon
komega_sst
r/R
Mean velocity
0.0
0.2
0.4
0.6
0.8
1.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

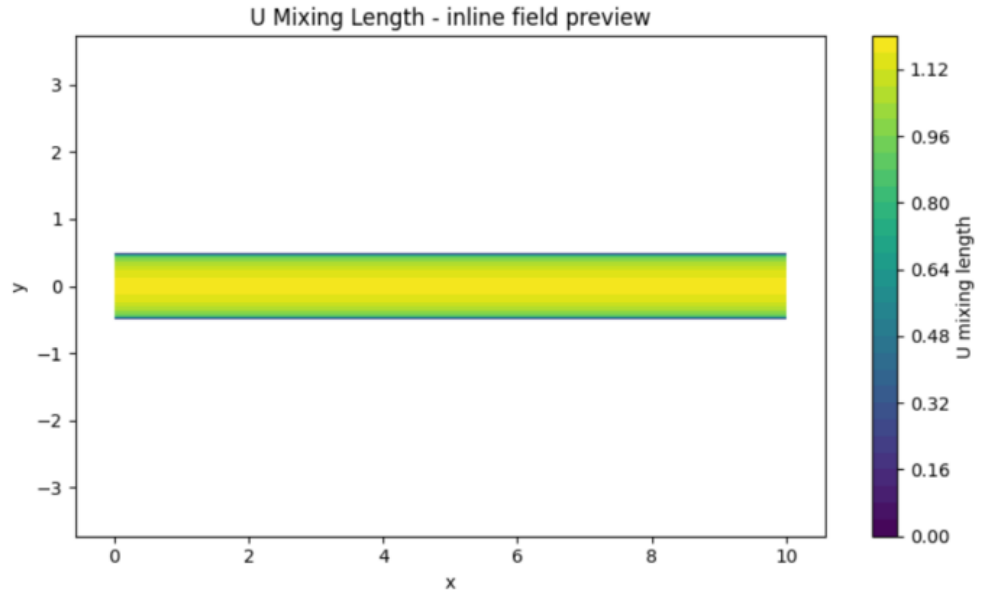

Friction-factor comparison
kepsilon
komega_sst
laminar
mixing_length
van_driest
Blasius
Haaland
Darcy friction factor
Reynolds number
10^{-2}
10^{3}
10^{4}
10^{5}



For Raynolds number 10^5:




Friction-factor comparison
kepsilon
komega_sst
laminar
mixing_length
van_driest
Blasius
Haaland
Darcy friction factor
10^{-2}
10^{-3}
10^{4}
10^{5}
10^{6}
Reynolds number


U Mixing Length - inline field preview
y
x
U mixing length
1.12
0.96
0.80
0.64
0.48
0.32
0.16
0.00

Results:

Parameter	**Re = 10^4**	**Re = 10^5**
Blasius friction factor (reference)	0.0316	0.0178
Haaland friction factor, smooth pipe (reference)	0.0309	0.0178
Centreline velocity, U_cl (fullness U_b / U_cl): laminar	2.00 (0.50)	2.00 (0.50)
Mixing length	≈ 1.25 (0.80)	≈ 1.14 (0.88)
Van Driest	≈ 1.20 (0.83)	≈ 1.14 (0.88)
k–ε	≈ 1.165 (0.86)	≈ 1.115 (0.90)
k–ω / SST surrogate	≈ 1.20 (0.83)	≈ 1.08 (0.93)
Darcy friction factor (ratio to Blasius): laminar	≈ 0.0064 (0.20)	≈ 0.00064 (0.04)
Mixing length	≈ 0.083 (2.6)	≈ 0.021 (1.20)
Van Driest	≈ 0.038 (1.19)	≈ 0.019 (1.09)
k–ε	≈ 0.026 (0.81)	≈ 0.0165 (0.93)
k–ω / SST surrogate	≈ 0.014 (0.45)	≈ 0.0062 (0.35)
Wall shear and pressure drop	proportional to f at fixed U_b	proportional to f at fixed U_b

Conclusion:

Radial eddy-viscosity closures (laminar, mixing-length, Van-Driest-damped mixing-length, equilibrium k–ε and a k–ω/SST surrogate) were compared for fully developed turbulent pipe flow at a unit bulk velocity, with the pressure gradient adjusted to obtain that bulk velocity, at Re = 10^4 and 10^5. The laminar case recovers the exact Poiseuille solution, with a centreline velocity of 2.00 and a friction factor of 64/Re (≈ 0.0064 and ≈ 0.00064), which confirms that the momentum solver

and the friction-factor evaluation are correct. It is included as a reference, since the flow is not laminar at these Reynolds numbers.

Profile fullness: All turbulent closures give much fuller profiles than the parabolic laminar one, with a flat core and a steep drop in a thin layer near the wall. Using the ratio of bulk to centreline velocity as a measure of fullness (0.5 for laminar), the values at $Re = 10^4$ are about 0.80 (mixing length), 0.83 (Van Driest), 0.86 (k–ε) and 0.83 (k–ω/SST surrogate), so k–ε gives the fullest profile by this measure, although the k–ω profile has the flattest core and the sharpest near-wall drop. At $Re = 10^5$ the ratios are about 0.88, 0.88, 0.90 and 0.93, and the k–ω/SST surrogate gives the fullest profile. Profiles become fuller as Re increases for every closure. The mixing-length and Van Driest profiles are almost identical at $Re = 10^5$, because wall damping only affects a thin viscous layer that shrinks as Re rises. For comparison, the usual relation $U_{cl}/U_b \approx 1 + 1.43\sqrt{f}$ gives about 1.25 at $Re = 10^4$ and 1.19 at $Re = 10^5$ for smooth pipes. The mixing-length result at $Re = 10^4$ (≈ 1.25) matches this, while all other closures are slightly fuller than expected, and at $Re = 10^5$ all closures are fuller (1.08–1.14).

Friction factor against the references: The Blasius correlation gives 0.0316 and 0.0178, and the Haaland correlation 0.0309 and 0.0178, at $Re = 10^4$ and 10^5. The Van Driest closure is closest among the high-friction models, at about 19% above Blasius at $Re = 10^4$ and 9% above at $Re = 10^5$. The k–ε closure is slightly below the correlations (about 0.81 and 0.93 times Blasius). The undamped mixing-length closure over-predicts friction strongly, by a factor of about 2.6 at $Re = 10^4$ and 1.2 at $Re = 10^5$. The k–ω/SST surrogate under-predicts it severely, at only about 0.45 and 0.35 times Blasius, which is not physically acceptable for smooth-pipe flow. Because wall shear and pressure drop at fixed bulk velocity are both proportional to the friction factor, they follow the same ordering and the same errors.

Consistency of the results: The friction factor and the profile fullness do not agree for every closure. A mixing-length friction factor of 0.083 at $Re = 10^4$ would imply a centreline velocity of about 1.4 through the relation above, but the computed profile has about 1.25. Likewise, the k–ω/SST surrogate gives the fullest profile at $Re = 10^5$ but the lowest friction. This suggests that the wall shear used for the friction factor is sensitive to the near-wall treatment, for example the first-cell velocity gradient or a wall function, and not only to the bulk profile. The velocity profiles of the k–ε and k–ω closures also end at non-zero velocity at $r/R = 1$, which suggests that their first point lies outside the viscous sublayer, as in a wall-function approach. The friction values from these two closures should therefore be treated as less reliable than the profile shapes.

Limitations of the figures: The friction-factor plots show one Reynolds number each, and the Blasius and Haaland lines are listed in the legends but not drawn, so the comparison above uses the correlation values calculated from the formulas. The intermediate case at $Re \approx 3.16 \times 10^4$, the wall shear, the pressure drop and the maximum turbulent-viscosity ratio were requested in the observations but are not shown in the figures. The field plot is available only for the mixing-length

case at Re = 10^4, where the streamwise velocity is uniform along the pipe with a maximum of about 1.2, as expected for a fully developed flow.

Overall, every turbulent closure produces the expected fuller-than-laminar profile, and the profiles become fuller with Reynolds number. The Van Driest mixing-length model gives friction factors closest to the Blasius and Haaland correlations (within about 20%), k–ε is slightly low, the plain mixing-length model is too high at Re = 10^4, and the k–ω/SST surrogate under-predicts the friction by more than half. Fullness alone is therefore not a measure of correctness: the closure with the fullest profile at Re = 10^5 also has the least accurate friction factor. Plotting the correlations, showing the Re ≈ 3.16×10^4 case, checking the near-wall resolution and wall treatment (y^+ of the first cell), and reporting the turbulent-viscosity ratio would complete the comparison.